Huang YuTing

EDUCATION

Lingnan University (Hong Kong) Sep 2025 - Present

Artificial Intelligence and Business Analytics Master

- GPA: 3.87 / 4.00
- Main course: Business Data Management, Generative AI, Big Data Analytics

Tzu Chi University Sep 2021

Master of science in Medical Informatics

- GPA: 4.00 / 4.00 (top 1 in the department)
- Research Project: Fusing Spatial and Channel Attention Mechanism in Reconstruction Residual Network for Diabetic Foot Ulcer Classification
- Main course: Deep Learning, Machine Learning, Software Engineering, Fuzzy Logic Theory and Applications, Medical Informatics

Tzu Chi University Sep 2017

Bachelor of science in Medical Informatics

- GPA: 3.68 / 4.00 (top 5 in the department)
- Related Courses: Introduction to Biomedical Informatics, Web programming, Molecular Biology, Database System, System Analysis and Design, Medicine and Information Ethics, etc.

RESEARCH EXPERIENCE

Toxicity Prediction using Graph Neural Networks Aug 2024 - Sep 2025

Researcher

- Built and evaluated multiple GNN architectures for toxicity prediction on public benchmark datasets.
- Enhanced model performance and interpretability by integrating atom-level, functional group, and fingerprint-based features with functional group importance analysis.

Microscopic image classification of Gramnegative bacteria Dec 2020 - Feb 2021

Researcher

- Description: The aim is to distinguish Gram-positive bacteria in microscopic images by CNN deep learning methods.
- Details: Design and train deep learning models, write model architectures and perform parameter and architecture tuning. The primary development language is python.
- Results: Using the SE-ResNet model, five different types of bacteria were identified with an accuracy of over 98%.

Hemodialysis Health Education App Dec 2019 - Dec 2020

Developer

- Description: The APP is designed to address the health education burden on clinicians and improve the quality of care for patients and families at home.
- Details: The whole app includes C# windows development and Android mobile device development. The development content includes system analysis, database design, app design, window programming, and data transfer.
- Result: The app was launched in Tzu-Chi Hospital and the satisfaction rate of patients and doctors reached 4.8/5.0.

PROFESSIONAL EXPERIENCE

Hong Kong Metropolitan University, Hong Kong | Science & Techonology Apr 2024 - Present

Research Assistant

- Description: Focused on applying deep learning models to address specific challenges in the field of bioinformatics, including the prediction of small molecule toxicity.
- Details: Data collection and organization, data cleaning, designing and writing model architectures.

Tzu Chi Hospital, Hualien, Taiwan | Hemodialysis Unit, Nursing Department Dec 2019 - Feb 2022

Programmer

- Description： Participate in the development of the hemodialysis room health education APP, develop a diabetic foot wound classification prediction project, complete demand research, function positioning, code writing, and technical support work.
- Detail: To understand the current situation of the industry, explore the needs of the hospital, and solve technical problems.

Zhangzhou Municipal Hospital, China | Information Department Jun 2020 - Sep 2020

Intern

- Description: Mainly involved in the maintenance of hospital information systems, including the design and production of nursing notes and assessment sheets, handling computer faults, and database data processing.
- Details: Complete the complete closed loop of the product cycle including community hospital information system design, requirement orientation, and system architecture design, and make recommendations for the development of hospital information technology.

Zhangzhou Third Hospital, China | Information Department Jul 2019 - Aug 2019

Intern

- Description: Hardware maintenance, troubleshooting, network cable creation, SQL syntax for database system queries, and learning hospital information system workflow.
- Details: Gained hands-on experience in the hospital information system, understanding workflows related to patient management, scheduling, and data entry.

EXTRACURRICULAR ACTIVITIES

Tzu Chi University | Department of Medical Information Sep 2021 - Present

- **Website maintainer:** independent maintenance of the website, including updating announcements, rotations, collation and updating of page information, design, and revision of website layout, etc.
- **Teaching Assistant:** Assist teachers with after-school teaching and learning, answering questions and providing guidance on assignments, providing timely feedback to teachers on students' problems, and setting directions for teaching priorities.

Tzu Chi University Sep 2018 - Sep 2021

Student Group Organisers

- **Club management:** Serving as the core officer of the International Students' Association and the officer of the Departmental Students' Association, managing the day-to-day work of the club, assisting in organizing various student activities and celebrations, and being able to host events with 300+ participants independently.
- **Event planning:** Planning and organizing school events, including seminars, freshers' camps, sports days, graduation ceremonies, etc., winning more than 10 awards.

ACHIEVEMENTS

Publications

[1] **Huang YT.**, Peng DL., Wang DD. *Learning of Molecular Graphs in Toxicity Prediction.* International Conference on Machine Learning and Cybernetics(ICMLC) (2024).

[2] Wang, JG., **Huang YT.** *Reconstruction residual network with a fused spatial-channel attention mechanism for automatically classifying diabetic foot ulcer.* Phys Eng Sci Med 47, 1581–1592 (2024).

[3] Yi YQ., **Huang YT.**, Wan X., Bian YT., Wang DD., Zhao PL., Ou-Yang L. *ETDock: An Equivariant Transformer for Protein-Ligand Docking.* Frontiers of Computer Science (Accepted).

[4] **Huang YT.**, Wang DD. *Substructure-guided Deep Graph Learning in Molecular Toxicity Prediction.* IEEE Journal of Biomedical and Health Informatics (Under Revision).

Project Awards

- Best Popularity Award - Hemodialysis Health Education APP

Campus Honors

- Outstanding Youth of Tzu Chi University
- Outstanding Graduates of Tzu Chi University

MISCELLANEOUS

- **Skills:** Programming Language: python, c, html, SQL; Software Framework: Visual Studio, VScode, SQL Server, Tensorflow+Keras
- **Certifications:** Microsoft AI900; QTC+ Python3
- **Languages:** Mandarine-Native; English; Taiwanese; Cantonese
- **Activities:** Hong Kong Techathon+(2025), ICMLC(2024), Medical Informatics Taiwan Competition (2022), Taiwan Research Institute of Industry AIGO problem solving competition (2022)